LESSON PLAN

NDT (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Ice protection, hot air protection, electrical system Calculation of engine thrust, various parameters and tests, adjustments of engine cont
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Ice protection, hot air protection, electrical system Calculation of engine thrust, various parameters and tests, adjustments of engine controls
5. Specific Objectives	20.1.2.1 Trainee will be able to understand and appreciate noise separation methods, thrust reversal mechanism, after burning fuel system, lubrication system, thrust augmentation devices, cooling and thrust calculations

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine control adjustments and parameter verification procedures; Safety: Ensure no live engine operation; Supervision: Instructor monitors theoretical understanding only
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary methods used for ice protection on aircraft?
(b) How does the thrust reversal mechanism function during landing?
(c) Which electrical systems are critical for engine thrust calculation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Ice protection and hot air systems • Thrust reversal and afterburning fuel systems • Lubrication and thrust augmentation devices	L & Ds	PPT, IB	15 min
(b)	• Calculation of engine thrust parameters • Engine cooling system analysis • Noise separation methods overview	L & Ds	PPT, PDF	20 min
(c)	• Various engine parameter tests • Adjustment of engine controls • Verification of thrust augmentation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between thrust augmentation devices and engine cooling requirements.
(b) Describe the steps involved in adjusting engine controls based on calculated thrust parameters.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts learned in this lesson apply to the engine control procedures covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min